



Exercise 1

Comprehension is a useful tool for generating artificial data.

1. Use a list comprehension to create a vector containing all numbers of the form x^y , with $x = 1, \dots, 10$; $y = 2, \dots, 7$; and $y > x$.
2. Compute the sum of the square roots of these numbers.

Exercise 2

Writing a simple function

This short optional exercise practices writing functions and loops.

1. Write a function that returns the smallest entry of a vector, without using built-in functions such as minimum or min.
2. Test your function on a vector of 10 randomly sampled integers in the range 1 to 100.

Exercise 3

A common trick to generate 100 samples of a binary variable with probability 0.7 for True (and 0.3 for False) is:

```
import numpy as np
np.random.rand(100) <= 0.7
```

Explain why this trick produces the desired result.